

Spatial Storytelling in the Newsroom: Reconstructing News Events in 3D

Malika Khurana
The New York Times, USA
malika.khurana@nytimes.com

Nick Bartzokas
The New York Times, USA
nicholas.bartzokas@nytimes.com

Karthik Patanjali
The New York Times, USA
karthik.patanjali@nytimes.com

ABSTRACT

When we create stories in 3-D space, we can go beyond words to help the reader feel they were there to experience and understand it. This talk shows how three visual stories in the New York Times create a sense of presence, using computer vision techniques, spatial audio, and 3-D tiles.

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1 INTRODUCTION

An airborne virus that spreads through particles and droplets was a perfect candidate for 3-D and spatial analysis, and the subject matter allowed us to present the data in its true form using immersive storytelling on multiple platforms. Through WebGL and augmented reality, these stories demonstrated how infectious aerosols can move in spaces of vital interest to public health: within the fibers of a face mask and in a classroom.

2 PUBLISHED ARTICLES

The following are the published articles we'll be presenting in our session:

2.1 Inside the Deadly Maui Inferno, Hour by Hour

The devastating fire in Maui in August 2023 spread over the course of hours and trapped over 100 people. Our team conducted a spatial investigation to understand what happened, piecing together videos from social media and eyewitness accounts.

Using Google 3-D Tiles and CesiumJS, we were able to present our findings by smoothly moving from an on-the-ground perspective, where we could show what happened through specific people's eyes, to an omniscient map-level perspective, where we could pull back and show how these pieces fit together [Baker et al., 2023].

2.2 The Quest to Restore Notre Dame's Glorious Sound

We wanted to take readers into an immersive audio experience to understand the efforts to restore the unique acoustics of the Notre Dame Cathedral [Schwartz et al., 2023], which were deeply affected by the devastating fire in 2019. The sonic landscape of the cathedral, once described by Victor Hugo as being "fraught with such benediction and such majesty, that they soothed this ailing soul" was an important element that scientists are striving to bring back to life.

Researchers, who had made detailed acoustic measurements of the cathedral in 2015, used these measurements to create a computer model that simulates the cathedral's acoustics. This model allowed the team to understand the potential impact of proposed changes on the cathedral's sound.

We worked with this research team and used the acoustic model to give our readers a first person visual and spatial audio experience of what different spots inside the cathedral sound like. The piece takes the reader through different parts of the cathedral and lets them hear spatially-accurate ambisonic audio that is accurately rendered based on the direction the virtual camera is pointed at as they scroll.

We wanted to interactively convey the fact that the sound they hear is not in isolation but entangled with the space itself. Incredible 3D models of the interior combined with a custom ambisonic spatial audio player that was developed in-house, helped us realize our creative vision and tell the story about sound in its most natural form.

Three primary data sets were used to create the virtual reconstruction of the concert presented in this article. The first was from the detailed acoustic and spatial measurements taken inside Notre Dame by Brian Katz in 2015. The second was a 3-D model of the interior of the cathedral. And the third was the audio recording of the performance of "Viderunt Omnes" in an echoless chamber. The 3-D model was integrated with the acoustic data using specialized software to simulate how sound propagates in the cathedral. These results were combined with the audio recordings to simulate how the choir would have sounded in different parts of the cathedral and encoded in a spatial audio format.

We wanted our readers to experience this spatial audio, but this traditionally requires special headphones that track the user's head movement to give that spatial effect. Since we wanted every reader to experience it, we built a custom spatial audio player that can mix the audio in real time based on the orientation and movement of a virtual camera controlled by scrolling the web page. This means that no special headphone hardware or software was needed to experience this 360° soundscape. It's accessible to any user with an internet connection and basic headphones.

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2.3 FIFA World Cup and Super Bowl coverage

We wanted to offer our readers an immersive view into key moments of the 2022 World Cup [Cai et al., 2022]. Through innovative storytelling and a custom, in-house production pipeline that used artificial intelligence techniques, we were able to deliver an immersive story within a few hours of the game, something that would have taken us days and weeks to produce in the past.

Our mission was to go beyond traditional visual coverage and create an interactive experience that stopped the action at strategically significant moments. We wanted to explore the intricate aspects of soccer, highlighting crucial player movements and action away from the ball.

To achieve this, our pieces did the following:

We slowed down the action and guided readers through critical moments, allowing them to absorb the game. We illustrated the dynamic movement of the ball and players away from the ball simultaneously, showcasing the interplay between the two. We provided an immersive view into the perspective of players, enabling readers to understand their decision-making process, passing lanes, and runs made by their teammates. We delivered this immersive analysis within an incredibly short timeframe of four to five hours from the final whistle, necessitating the development of new tools and a streamlined production pipeline.

3 CONCLUSION

These projects demonstrate the application of computer vision techniques, spatial audio, and 3-D tiles in custom workflows that enable the New York Times to augment our reporting beyond traditional media. Because we integrate these applied techniques in a smooth storytelling experience on the web, readers can focus on the spatial relationships that are core to understanding certain news events and phenomena, without getting distracted by the technology itself. In many cases, we create 3-D stories on tight timelines, opening up the possibility for spatial analysis of breaking news events hours after they happen.

Notes: The 3-D model above is an illustration of the cathedral's interior before the fire. It was developed by Sunmetron Architects. Lidar 3-D scans from the Andrew Tallon Papers were used for adding detail. The calibrated acoustic model of Notre Dame was originally created by Brian F.G. Katz and Bart N.J. Postma, with historical modifications and audio production by Sarabeth S. Mullins, Elliot K. Canfield-Dafilou and David Poirier-Quinot. Audio samples are of the Ensemble Rue des Chantres: Christian Ploix, Erwan Picquet, Vincent Pislár and Renaud Tripathi.

Methodology: Three primary data sets were used to create the virtual reconstruction of the concert presented in this article. The first was from the detailed acoustic and spatial measurements taken inside Notre Dame by Brian Katz in 2015. The second was a 3-D model of the interior of the cathedral. And the third was the audio recording of the performance of "Viderunt Omnes" in an echoless chamber. The 3-D model was integrated with the acoustic data using the CATT-Acoustic/TUCT software to simulate how sound propagates in the cathedral. These results were combined with the audio recordings to simulate how the choir would have sounded in different parts of the cathedral. Sources: Institut Jean Le Rond d'Alembert, a research laboratory directed by Pierre-Yves

Lagrée, which draws from the C.N.R.S. and the faculty of sciences and engineering at the Sorbonne University. 3-D scans provided courtesy of the Vassar College Library.

The pre-fire 3-D base map of Lahaina visualized in this article uses Google's Photorealistic 3D Tiles. Google incorporates the following sources to create the tiles: The General Bathymetric Chart of the Oceans; The International Bathymetric Chart of the Arctic Ocean; Columbia University's Lamont-Doherty Earth Observatory; Landsat; Copernicus; Monterey Bay Aquarium Research Institute; Airbus; National Geospatial-Intelligence Agency; National Oceanic and Atmospheric Administration; National Science Foundation; Scripps Institution of Oceanography; School of Ocean and Earth Science and Technology, University of Hawai'i at Mānoa; U.S. Navy and United States Geological Survey. The Times used CesiumJS, a Javascript library for visualizing 3-D maps, to display the base map.

The 3-D scene for the Soccer World Cup articles were created using photographs and videos of the venue for reference. The positions of the players were determined from a photograph taken at that moment during the play. The player models were created using a machine learning model called ICON, which was created by Yuliang Xiu, Jinlong Yang, Dimitrios Tzionas, Michael J. Black and the Max Planck Institute. The models were refined manually using photographs as reference.

REFERENCES

- Mike Baker, Malika Khurana, K.K. Rebecca Lai, Riley Mellen, Natalie Reneau, Bedel Saget, Elena Shao, Anjali Singhvi and Charlie Smart. 2023. The New York Times. Inside the Deadly Maui Inferno, Hour by Hour. <https://www.nytimes.com/interactive/2023/11/01/us/hawaii-maui-fire-timeline.html> </bib>
- Madeleine Schwartz, Malika Khurana, Mika Gröndahl and Yuliya Parshina-Kottas. 2023. The New York Times. The Quest to Restore Notre Dame's Glorious Sound. <https://www.nytimes.com/interactive/2023/03/03/magazine/notre-dame-cathedral-acoustics-sound.html> </bib>
- Weiwei Cai, Chloé Desaulles, Alexandre Devaux, Or Fleisher, Malika Khurana, Allison McCann, Mark McKeague and Bedel Saget. 2022. The New York Times. Watch Christian Pulisic Send the U.S. Past Iran and Into the Knockout Rounds. <https://www.nytimes.com/interactive/2022/11/29/sports/world-cup/iran-usa-world-cup-goal-pulisic.html>